

Review

A comprehensive review of large language models: issues and solutions in learning environments

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Abstract

A significant advancement in artificial intelligence is the development of large language models (LLMs). Despite opposition and explicit bans by some authorities, LLMs continue to play a transformative role, particularly in education, by improving language understanding and generation capabilities. This study explores LLMs' types, history, and training processes, alongside their application in education, including digital and higher education settings. A novel theoretical framework is proposed to guide the integration of LLMs into education, addressing key challenges such as personalization, ethical concerns, and adaptability. Furthermore, the study presents practical case studies and solutions to barriers, such as data privacy and bias, offering insights into their role in enhancing the teaching–learning process. By providing a systematic analysis and proposing a structured framework, this study advances current knowledge and highlights the significant potential of LLMs in revolutionizing education.

Keywords Natural language processing systems · Large language models · Neural networks · Artificial intelligence · Education · Learning systems

1 Introduction

1.1 Context

Artificial intelligence and LLMs are now widely used in every field. Their powerful language understanding and computational capabilities open the door to innovative AI applications. A notable example is ChatGPT-4, which represents the state-of-the-art in LLM development [1]. These models rely on deep learning and extensive training with big data to analyze patterns in human language, respond to questions, and perform complex tasks [2, 3].

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The education sector is experiencing significant transformations, as LLMs offer opportunities to automate processes like content creation, grading, and personalized learning [4–7]. However, these advancements also raise critical ethical and practical challenges, such as data privacy, bias, overreliance on AI, and pedagogical concerns [8–10]. Addressing these challenges requires a structured approach to ensure that LLMs enhance, rather than hinder, educational systems. Figure 1 illustrates the architecture of LLM in education.

Several existing studies have explored the technical capabilities and ethical implications of LLMs [8, 11], yet critical gaps remain. Specifically, there is a lack of theoretical frameworks that outline how LLMs can be effectively integrated into education while addressing personalization, scalability, and ethical concerns. Additionally, previous studies tend to focus on individual applications rather than providing a systematic analysis of real-world educational use cases [12, 13]. Figure 2 illustrates the building block of LLMs.

To address these limitations, this study makes the following contributions:

- Proposing a novel theoretical framework for integrating LLMs in education, focusing on three key pillars: personalized learning, ethical and pedagogical balance, and learning adaptability.
- Conducting a systematic review of LLM capabilities, training processes, and applications in educational settings [14, 15].
- Providing practical case studies to illustrate real-world challenges, opportunities, and solutions in implementing LLMs.

By systematically bridging the existing gaps and offering a structured framework, this study highlights the potential of LLMs to transform teaching and learning while addressing key challenges.

1.2 Motivation of the study

This study is motivated by the transformative impact of ChatGPT and LLMs across disciplines, particularly in education, healthcare, finance, coding, and the job market.

In education, LLMs like ChatGPT revolutionize learning by virtualizing processes, expanding opportunities, and reshaping pedagogical approaches. Their potential to automate tasks, enhance personalized learning, and improve accessibility highlights their significance.

Other sectors also demonstrate their impact:

- Healthcare: Assisting in medical diagnosis, research, and knowledge dissemination [8].

Fig. 1 Architecture of LLMs for education

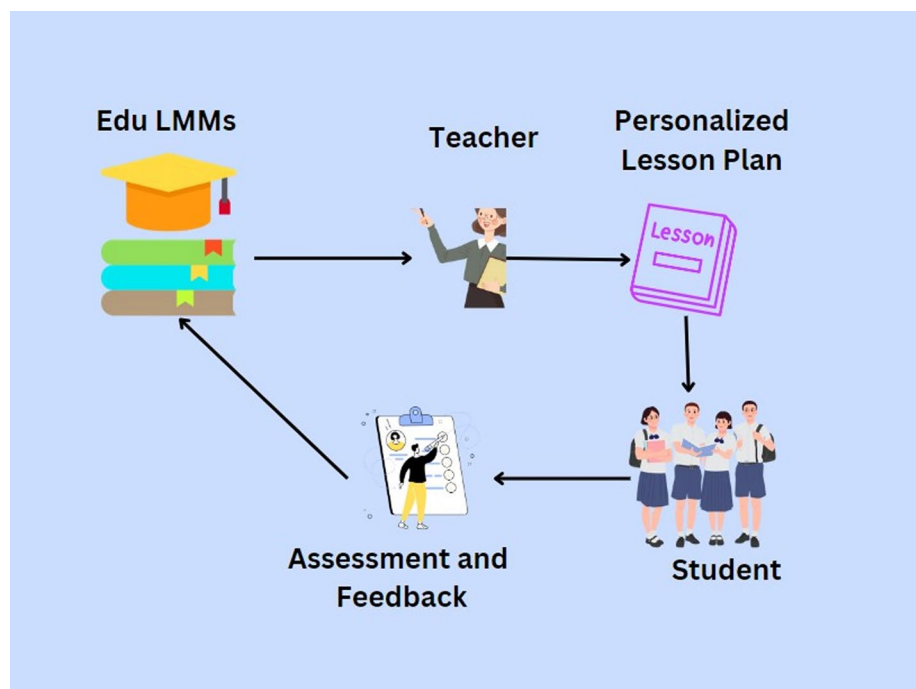
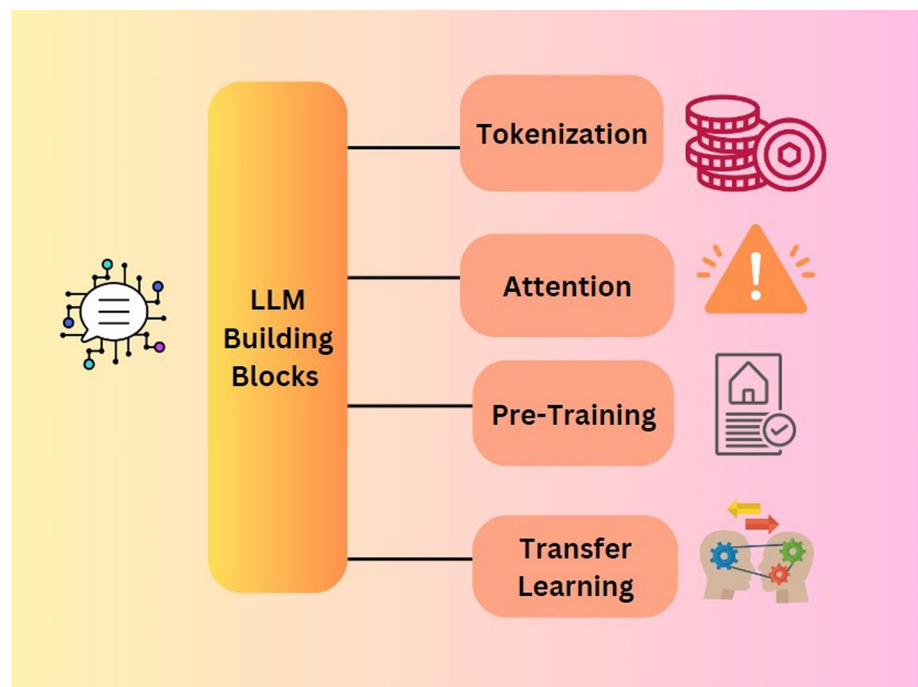


Fig. 2 LLMs buildings blocks

- Finance: Supporting financial analysis, market predictions, and investment guidance [3].
- Coding Proficiency: Improving code generation, debugging, and enhancing developer productivity [6].
- Labor Market: Digitizing labor processes and reshaping job requirements [7].

Despite these advancements, challenges remain, such as understanding the ethical, practical, and technical implications of integrating LLMs. This study focuses on educational applications of LLMs, evaluating their capabilities, addressing challenges, and proposing solutions to optimize their role in teaching, learning, and administrative tasks.

1.3 Contribution of the study

The main contributions of this article are as follows:

- Comprehensive Background and Theoretical Foundation: The study begins by providing an extensive background on the evolution of educational systems, LLMs, and the concept of intelligent education. It clarifies the relationship between LLMs and education and explores the notion of “Smart Education” as a transformative paradigm, setting the stage for the practical and theoretical insights presented.
- Novel Theoretical Framework for LLM Integration: A proposed theoretical framework is introduced to guide the ethical and effective integration of LLMs, such as ChatGPT, into educational systems. The framework provides actionable guidance by focusing on three interdependent pillars:
 - o Personalized Learning Models: Offering tailored educational content to meet individual student needs through adaptive systems.
 - o Ethical and Pedagogical Balance: Developing strategies to address challenges like AI bias, data privacy, and over-reliance on AI, while enhancing critical thinking and human oversight.
 - o Learning Adaptability Framework: Ensuring LLM flexibility across diverse educational contexts, such as K-12 classrooms, higher education, and online learning.
- Systematic Review of LLMs: The article conducts a structured and comprehensive review of LLMs, including their history, training processes, and evaluation methods. This systematic review identifies underexplored areas in educational applications, such as the need for bias mitigation, privacy protections, and energy-efficient deployments.

- **Comparative Analysis with Practical Insights:** To ensure actionable implications, the study presents real-world case studies demonstrating LLM applications in education. These include:
 - o Automated content creation: Reducing the workload of educators.
 - o Intelligent tutoring systems: Providing students with personalized feedback and tailored learning experiences.
 - o Language support tools: Enhancing accessibility for non-native speakers.
 - o A comparative analysis evaluates these applications across primary education, higher education, and digital learning, offering actionable insights for educators, policymakers, and institutions.
- **Challenges and Actionable Solutions:** The study identifies key challenges in the adoption of LLMs, including data privacy, ethical considerations, algorithmic bias, and pedagogical limitations. To address these challenges, the article proposes practical solutions, such as:
 - o Implementing robust data protection frameworks (e.g., encryption, anonymization).
 - o Developing bias mitigation strategies to ensure fairness and inclusivity.
 - o Promoting teacher-AI collaboration to balance automation with human oversight.
 - o Encouraging sustainable deployment of LLMs with energy-efficient practices.
- **Practical Recommendations for Higher Education:** The study highlights actionable recommendations for implementing LLMs in higher education, including:
 - o Enhancing personalized feedback mechanisms to improve student learning outcomes.
 - o Supporting remote learning environments through AI-powered tools.
 - o Addressing academic integrity concerns with ethical monitoring frameworks to ensure responsible LLM use.

By combining a novel theoretical framework, a systematic review, and practical case studies, this article provides both theoretical advancements and actionable recommendations. It bridges existing research gaps and offers concrete solutions for educators, policymakers, and researchers to guide the ethical and effective integration of LLMs in education.

1.4 Organization of paper

The remaining sections of the paper follow this structure: Section II explains the research methodology and the research methodology, in which protocol, research questions, keyword selections, paper selections, formulation of strings, database selections and inclusion and exclusion criteria are discussed. Section III explains the results of the research questions and Section IV concludes the conclusion and future work. A list of abbreviations is given in Table 1.

Table 1 List of abbreviations

Abbreviation	Description	Abbreviation	Description
LM	Large Language Model	AI	Artificial Intelligence
NLP	Natural Language Processing	DL	Deep Learning
NN	Neural Network	LM	Language Model
RNN	Recurrent Neural Network	RNNLM	Recurrent Neural Network Language Model
GPT	Generative Pre-trained Transformer	VAEs	Variable Auto encoders
HEIs	Higher Education Institutions	BERT	Bidirectional Encoder Representations from Trans-formers
MLM	Multi-layer masking	NSP	Next Sentence Prediction

2 Literature review

LLMs are significant in multidisciplinary areas like artificial intelligence and data science. They play a significant role in AI research because they try to imitate human language skills and find solutions for issues with NLP. We can achieve this through techniques based on deep learning and training with large volumes of data [16]. These LLMs are widely used in the field of data science. Their applications include machine translation, sentiment analysis, text mining [17], and extracting relevant information from textual data. LLMs also have connections to cognitive science, machine learning (ML), and computer science, which encourages interdisciplinary research and cooperation [18].

LLMs like GPT-3 have recently been introduced, and many people are interested in them and have strong opinions about their transformative potential. Deep learning-powered LLMs have excellent language generation and understanding capabilities, significantly advancing the field of artificial intelligence [18]. These advances are primarily driven by architectures like Transformers, which rely on self-attention mechanisms to process sequential data efficiently [3, 6]. Pre-training on vast amounts of textual data enables LLMs to perform well on downstream tasks like sentiment analysis, translation, and summarization [14, 15, 19].

The education sector faces a multitude of opportunities and challenges. Traditional teaching struggles with issues like differentiated education, unequal access to resources, and effectiveness measurement [20]. Advances in EduLLMs can address these challenges by offering tailored learning materials, personalized feedback, and improved engagement [7, 10, 12]. However, practical implementation raises concerns about ethical usage, data privacy, and algorithmic bias [8, 9, 21, 22]. Addressing these concerns is crucial to leveraging LLMs in education effectively.

Figure 3 illustrates the significance and several steps to the development of LLMs.

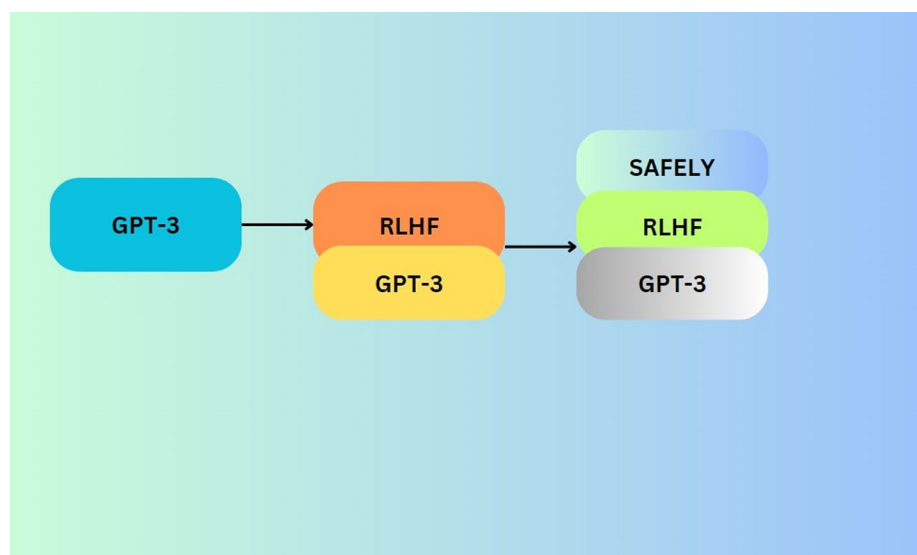
2.1 Applications of LLMs in education

If teachers follow a prescribed curriculum, traditional learning will remain the most common and publicly recognized form of education [23]. Conversely, the widespread use of digital technology has given rise to new paradigms in education. Online courses, distance learning, and digital tools provide alternatives to traditional classrooms [24], enabling personalized and independent learning.

EduLLMs (educational LLMs) are fundamental to these paradigms, as they enable tasks like semantic understanding, sentiment analysis, and text comprehension [3, 25]. By analyzing large datasets, EduLLMs can provide real-time question answering and extract meaningful insights to aid learning. Deep learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) further enhance their output quality [26, 27]. Hybrid deep learning techniques have been used to analyze student academic performance [28] as well as exam supervision [29].

Incorporating Reinforcement Learning into EduLLMs enables models to adapt to student feedback dynamically, improving performance over time [30]. RL models can optimize recommendations, train learning agents, and

Fig. 3 Steps of evolution Towards ChatGPT



automate responses to student inputs [19]. Additionally, Data Mining (DM) techniques help identify learning patterns, behavioral trends, and knowledge gaps, facilitating personalized learning environments [31, 32].

To improve engagement, EduLLMs integrate multimodal technologies, such as:

- Computer Vision (CV): Analyzing students' facial expressions and body language for real-time feedback [42].
- Voice Recognition: Supporting oral practice, speech evaluation, and pronunciation correction in language learning environments [33].

These technologies allow EduLLMs to provide a flexible, real-time learning experience tailored to individual student needs. Chatbots and intelligent tutoring systems powered by LLMs offer 24/7 personalized assistance, reducing educators' workload and enhancing learning efficiency [34–36].

However, challenges persist regarding the ethical deployment of LLMs in education. Issues like data bias, student privacy, and algorithm transparency must be addressed to ensure equitable and inclusive learning experiences [9, 37, 38].

2.2 Evolution of LLMs

The evolution of LLMs demonstrates their increasing capacity to handle complex NLP tasks. Early models like GPT-1 showed flexibility in tasks such as language translation, sentiment analysis, and text categorization [3].

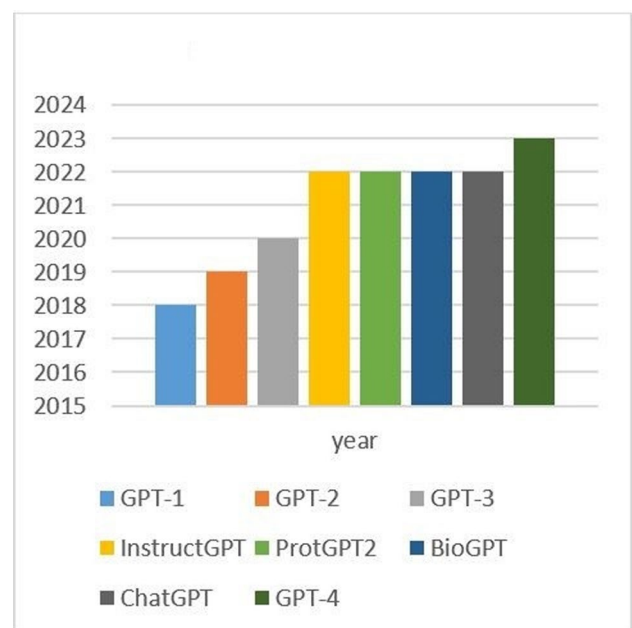
The release of GPT-2 marked a turning point, as it introduced significant enhancements over GPT-1. GPT-2, with 1.5 billion parameters, generated longer and more coherent text sequences, showcasing the ability to adapt to various domains [7, 39]. GPT-2's versatility enabled downstream tasks, including text summarization, classification, and question answering [40, 41].

Building on GPT-2, GPT-3 significantly expanded the parameter count to 175 billion, leading to substantial improvements in text generation accuracy and logical consistency [42]. GPT-3's architecture also introduced transfer learning strategies, where pre-trained models were fine-tuned for specific tasks with minimal labeled data [19].

The development of Instruct GPT further advanced GPT models by integrating reinforcement learning with human feedback (RLHF). Unlike GPT-3, Instruct GPT uses smaller, curated datasets to refine its outputs iteratively, improving reliability and alignment with user goals [43–45].

Figure 4 highlights the evolution of LLMs, showcasing their advancements in parameters, training processes, and task performance.

Fig. 4 Evolution of LLMs



2.3 Proposed theoretical framework

To address the challenges and gaps identified in existing literature regarding the integration of LLMs in education, this study proposes a novel theoretical framework. The framework is built on three interdependent pillars aimed at enhancing personalized learning, ethical AI integration, and adaptability within educational systems.

1. Personalized Learning Models
2. LLMs analyze student-specific data (e.g., learning pace, performance) to deliver tailored materials.
3. A dynamic feedback loop ensures continuous improvement in learning outcomes [10, 15].
4. Ethical and Pedagogical Balance
5. Ethical concerns, such as bias and overreliance, are addressed through transparent deployment strategies [21, 22].
6. Human-in-the-loop approaches ensure AI complements critical thinking instead of replacing it [38, 40].
7. Learning Adaptability Framework
8. The framework advocates for flexible AI systems capable of adapting to diverse educational environments, such as K-12 and higher education [46, 47].
9. Integration of multimodal learning tools like AR/VR ensures inclusivity and scalability [48].

2.4 Research gap

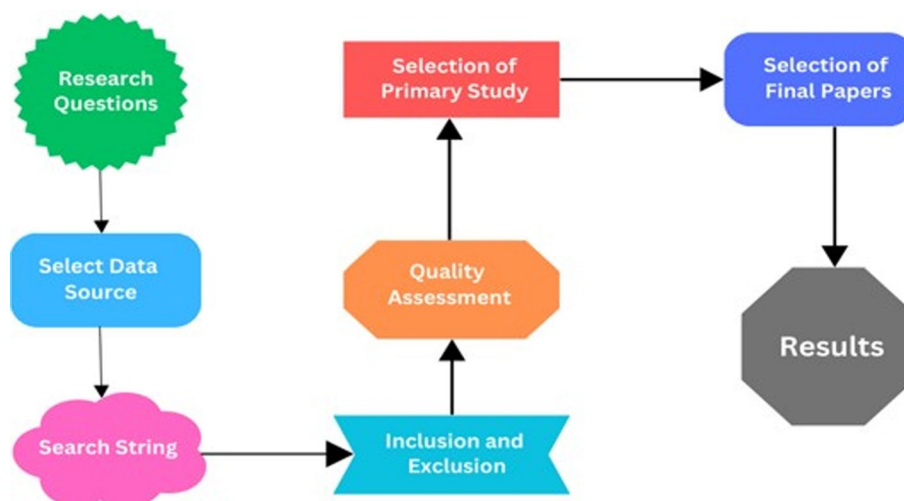
Despite the significant advancements in LLMs, their integration into educational systems remains fragmented, with several critical gaps in the current literature:

- Lack of Structured Frameworks for Integration: Existing studies primarily focus on isolated applications of LLMs, such as automated content creation, language tutoring, and intelligent tutoring systems [25, 34, 35]. However, these studies fail to provide a structured, theoretical framework to guide the integration of LLMs into education while addressing scalability, adaptability, and ethical challenges.
- Limited Research on Ethical and Pedagogical Challenges: While LLMs offer opportunities for personalized learning, significant concerns remain about data privacy, algorithmic bias, and ethical deployment [9, 22, 37]. Current literature lacks a comprehensive approach to balancing the pedagogical benefits of LLMs with the need to ensure ethical AI usage in education.
- Insufficient Exploration of Multi-Contextual Adaptability: Most existing studies focus on specific educational settings (e.g., higher education) or particular tasks (e.g., grading automation or question answering) [7, 10, 12]. There is limited exploration of how LLMs can adapt to diverse learning environments, including K-12 education, distance learning, and blended classrooms.
- Absence of Comparative Analysis and Real-World Applications: Few studies offer real-world case studies or comparative analyses of LLM implementations across educational contexts. This limits the practical understanding of how LLMs can address challenges like student engagement, learning personalization, and content accessibility in varying academic settings [6, 8, 36].
- Addressing Educational Inequities: While LLMs have the potential to transform education, there is insufficient research on how these models can bridge educational inequities, such as access to high-quality learning resources for underprivileged or marginalized communities [10, 15, 21].

3 Methods and materials

3.1 Research protocol

The study protocol refers to a plan and method to guide the analysis process. It provides a detailed analysis process, including the research questions to be addressed, a search strategy to identify relevant studies, inclusion and

Fig. 5 Flow of work

exclusion criteria for study selection, methods for data extraction and synthesis, and methods for assessing the quality of selected studies Fig. 5 illustrates the proposed methodology.

3.2 Planning the review

This phase lays the foundation for the systematic and structured approach of the SLR. During this phase, comprehensive guidelines are established to select the primary studies. By carefully planning the review, researchers can ensure that the review process is well-structured, systematic, and methodologically sound, leading to reliable and valuable insights. The critical aspects of this phase are defined below.

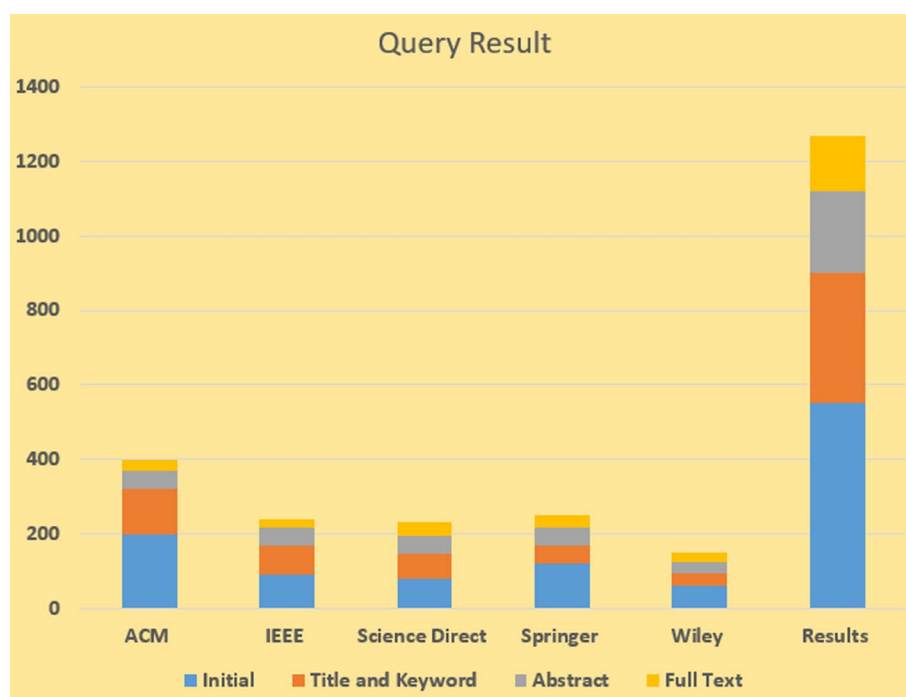
3.3 Identification of the research questions

The primary objective of this study is to conduct an SLR that identifies, analyzes, and summarizes empirical evidence related to the use of LLM in education. The review focuses on using LLM in education for teachers, students, and particular types of learners.

It also focuses on issues and their solutions for LLM in education. The research questions and the motivation behind each question have been formulated to guide the review process to achieve this goal. Table 2 illustrates the research questions.

Table 2 Research questions and motivation

Research questions	Motivation
RQ1: What is the history of LLMs?	To know about the history of LLMs and evaluation of the LLMs model
RQ2: What is the training and working process of LLMs?	To know about the model of LLMs, their training process, and working
RQ3: What is the role of LLMs in education and digital learning	To know about the role of LLMs in education and the digital learning process
RQ4: What is the process of implementing LLMs in higher education?	To know about LLMs' opportunities, challenges, barriers, and mitigation techniques in higher education
RQ5: what are the real-time education applications based on LLMs	To know about some case studies of LLMs in education
RQ6: What are the Key challenges and risks of applying LLMs in education?	To know about the challenges related to the application of LLMs in education
RQ7: What are the solutions to the key challenges and risks of applying LLMs in education?	To know about different types of suggestions and methods that are used to prevent information

Fig. 6 Query results**Table 3** Query results from data sources

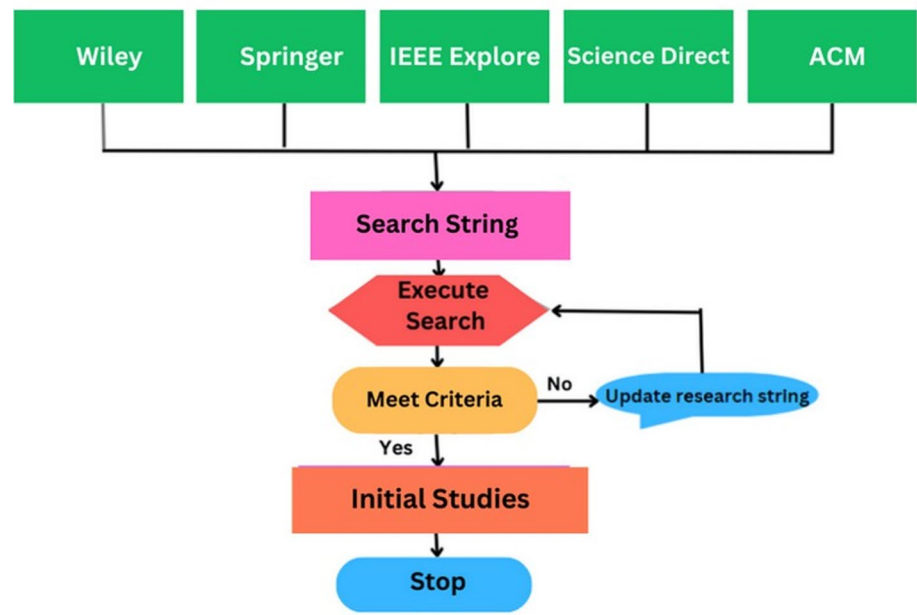
Data source	Initial	Title and Keywords	Abstract	Full Text Result
IEEE Xplore	90	80	45	25
Science Direct	80	65	50	35
ACM Digital Library,	200	120	50	30
Springer Link	120	50	45	35
Wiley	60	35	30	35
Results	550	350	220	150

3.4 Data sources selection

Data sources are the libraries from which the research studies should be retrieved. Four digital libraries have been chosen to extract the primary analyses. These digital libraries are IEEE Explore, Science Direct, the ACM Digital Library, and Springer Link [49]. The full text of the documents is searched to identify the prior studies. There are various options available to search each digital library for pertinent information. To find the most relevant literature, the search strategy is modified to satisfy the needs of the respective data source. Selected data sources and the number of studies produced by search queries are illustrated in Fig. 6 and Table 3.

3.5 Keywords searching

A combination of keywords and search operators is used to identify relevant studies that address the research question or topic of the review. This step focuses on specific keywords and their synonyms chosen from the identified research questions, as indicated in Table 2, to create the search string. These keywords are combined using the 'AND' and 'OR' conditions in the order listed to complete the following search string [50]. Figure 7 and Table 4 illustrate the process of formulating keywords.

Fig. 7 Keywords searching process**Table 4** Search string formulation

Keyword	Synonym/Alternative word
Education	("program" OR "system")
ChatGpt	("AI" OR "ML")
Application	("metrics" OR "classification")
Methods	("Techniques" OR "Framework")

3.6 Define inclusion and exclusion criteria

Inclusion and exclusion criteria are essential for ensuring objectivity and consistency in selecting studies for this systematic literature review (SLR). These criteria were carefully designed to align with the study's research objectives and ensure the quality and relevance of the included studies.

3.6.1 Inclusion criteria

The following inclusion criteria were applied to select studies:

- **Time Frame (2018–2023):** Studies published within this period were included to reflect recent advancements and trends in the use of LLMs in education. This ensures the review captures up-to-date research and emerging insights.
- **Language (English):** Only studies written in English were included to maintain consistency in interpretation and analysis.
- **Focus on LLMs in Education:** The selected studies must specifically address the application of LLMs, including ChatGPT, in the educational domain. This criterion ensures alignment with the research questions and objectives.
- **Empirical Research:** Studies involving practical experiments or empirical evaluations using specific datasets were prioritized to emphasize evidence-based findings.
- **Comprehensive Evaluation:** Each chosen study must provide a detailed evaluation of the performance, opportunities, and challenges of LLMs in education.
- **Source Quality:** The scope was limited to reputable journals, conferences, or books to ensure reliability and scholarly rigor.

3.6.2 Exclusion criteria

Studies meeting the following criteria were excluded from the review:

- **Published Before 2018:** Older studies were excluded to limit the scope to recent developments, given the rapid evolution of LLM technology.
- **Irrelevant Focus:** Studies that did not focus on LLMs or their applications in education were excluded to maintain relevance to the research questions.
- **Lack of Empirical Analysis:** Studies without empirical results, such as theoretical discussions or opinion pieces, were excluded to prioritize measurable insights.
- **Absence of Performance Evaluation:** Studies that failed to evaluate the performance or effectiveness of LLMs in education were excluded to ensure the findings contribute to actionable insights.

3.6.3 Justification for criteria

The inclusion and exclusion criteria were designed to ensure that the selected studies:

- Provide evidence-based insights relevant to the application of LLMs in education.
- Align with the study's focus on practical, empirical findings rather than theoretical or speculative discussions.
- Reflect recent developments and emerging trends, ensuring the review's relevance and significance.

3.6.4 Quality assessment

The selected studies were further evaluated using the Modified CASP Checklist to assess their methodological rigor and relevance. Table 5 illustrates the quality assessment criteria, focusing on critical aspects such as clarity of objectives, relevance to LLM applications in education, and empirical robustness.

These criteria ensured consistency and objectivity in study selection, enhancing the reliability of the review process.

3.7 Primary study selection

The primary studies selected for this review were curated using the tollgate approach, a structured five-phase methodology [50]. This approach ensured a systematic, transparent, and rigorous selection of studies aligned with the PRISMA guidelines.

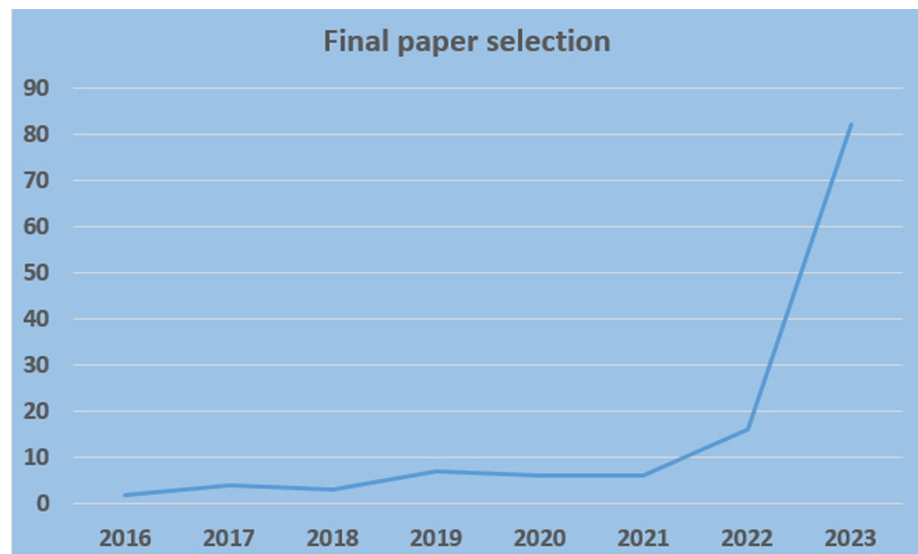
- **Phase 1 (Identification):** A comprehensive search was conducted across academic databases, including PubMed, IEEE Xplore, Scopus, and Web of Science, using specific keywords and Boolean operators. The search focused on studies published between 2016 and 2023 that explored LLM applications in education.
- **Phase 2 (Screening):** Duplicates were removed, and titles and abstracts were screened based on the following inclusion criteria:
 - Studies must focus on the application of LLMs (e.g., GPT models) in educational systems.
 - Articles must be peer-reviewed and written in English.
 - Only articles addressing LLM benefits, challenges, or implementation case studies were considered.

Table 5 Quality assessment criteria

Sr#	Quality assessment questions
C1	Does the study provide enough information about the history of LLMs?
C2	Does the study provide enough information about the model and training of LLMs?
C3	Does the study provide enough information about LLM's role in education?
C4	Does the study provide enough information about LLM's role in higher education?
C5	Does the study provide enough information about LLMs' challenges and their solutions in education?

Table 6 Final paper selection

Year	Final selection
2016	2
2017	4
2018	3
2019	7
2020	6
2021	6
2022	16
2023	82

Fig. 8 Final paper selection

- o Exclusion criteria included non-peer-reviewed work, opinion articles, and studies unrelated to educational applications.
- Phase 3 (Eligibility): Full-text screening of the remaining articles was conducted to ensure alignment with the research objectives. Reasons for exclusion (e.g., lack of relevance, insufficient methodological rigor) were recorded for transparency.
- Phase 4 (Quality Assessment): The quality of selected articles was evaluated using the Modified CASP Checklist, focusing on research design, clarity of objectives, validity, and relevance to the review. Articles scoring below a specified threshold were excluded. A table summarizing quality scores can be included in supplementary materials if needed.
- Phase 5 (Final Selection): Following quality assessment, 150 primary studies were selected for synthesis. Table 6 and Fig. 8 provide a breakdown of studies by year, and the PRISMA flow diagram (Fig. 9) illustrates the overall process.

3.8 Thematic analysis framework

To identify and synthesize recurring patterns and themes in the selected studies, this research employs the six-step thematic analysis framework proposed by Braun and Clarke [51]. This structured approach ensures a rigorous examination of qualitative data. The process began with an in-depth familiarization stage, where the full-text articles of all 150 primary studies were reviewed to gain a thorough understanding of their content. Repeated readings helped identify key concepts and patterns related to the research questions.

Following this, the data was systematically coded using qualitative data analysis software, focusing on identifying key phrases and ideas relevant to LLMs in education. Initial codes included concepts such as “bias mitigation,” “adaptive

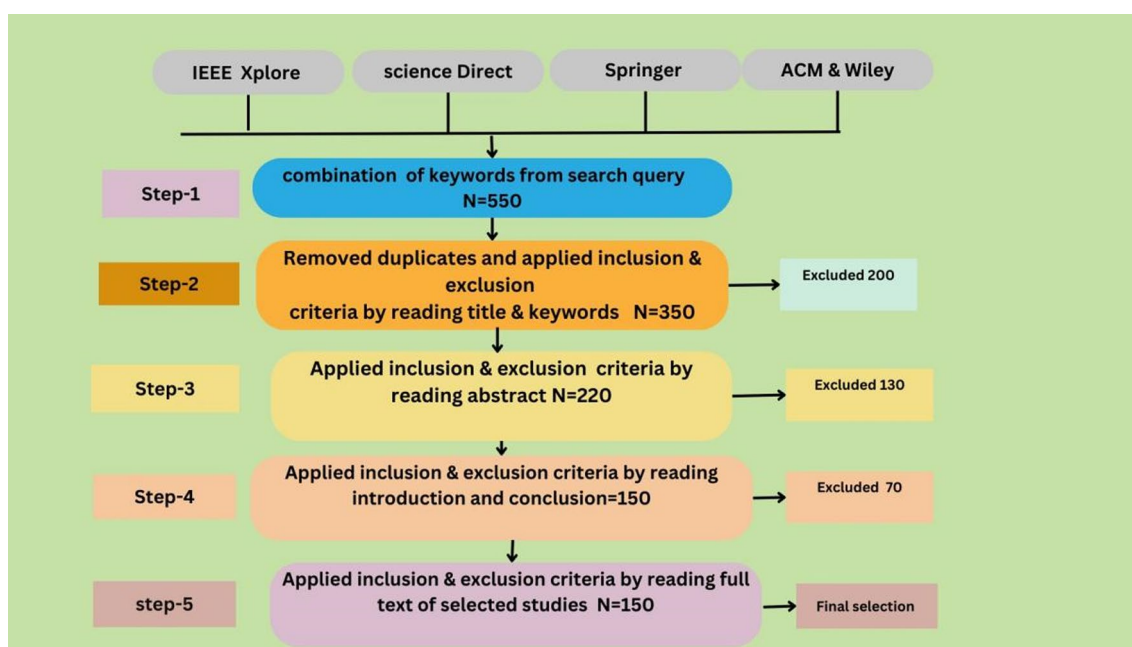


Fig. 9 PRISMA diagram

learning," "privacy concerns," and "teacher-AI collaboration." These codes were subsequently organized into broader themes by grouping related concepts. For instance, themes like "ethical dilemmas," "pedagogical impacts," and "cultural considerations" were identified as central to the research objectives.

The identified themes underwent iterative review to ensure consistency and alignment with the data. Overlapping themes, such as "bias detection" and "bias mitigation," were consolidated under the broader theme of "bias and fairness." Each theme was then clearly defined and scoped to capture its unique aspects. For example, the theme "cultural considerations" was defined as addressing the potential for LLMs to perpetuate dominant cultural norms while marginalizing diverse perspectives.

Finally, the themes were synthesized into actionable insights, which are presented in the Results (Sect. 4) and Discussion (Sect. 5) sections. This synthesis provides a structured narrative of the opportunities, challenges, and solutions related to integrating LLMs into education. By following this rigorous thematic analysis framework, the study ensures that its findings are both valid and methodologically sound.

4 Result

4.1 History of LLMs

LLMs are subsets of AI models [52] that are capable of deciphering and analyzing human-created language and using that knowledge to produce new text [53, 54]. LLMs are another term for language models. These models may be identified by examining large amounts of text [55]. The first LLM was developed in the 1950s and 1960s. The idea of LLM was initially put forward simultaneously. However, the models' developers needed help handling the complex natural language processing processes [56]. These models needed to improve their ability to understand English because of the rules and language characteristics that were put into them. These models also made use of language characteristics. The development of mathematical models reached its peak in the 1980s and 1990s. These models may calculate the probability of word sequences in a given context using probabilistic modeling approaches [57]. Even though ML algorithms have proven to be capable of effectively analyzing large datasets, there have been challenges in interpreting language, particularly contextual and semantic components [58].

The area of language modeling has grown significantly since the introduction of LLMs in the middle of the 2010s. The algorithms used deep learning techniques to evaluate vast amounts of textual data to identify structures and patterns in language usage [59]. A significant advancement was made in 2010 with the release of the recurrent neural network

language model, which produced clearer text by effectively predicting context [60]. Following the 2015 release of Google Neural Machine Translation, the first neural machine translation application used globally, improved performance on text samples from several countries was shown [33, 61].

In 2017, release of the Transformer model enabled language interactions and parallel training on many GPUs. In 2018, the OpenAI-created GPT-1 model demonstrated its ability to produce appropriate comments for the context.

GPT-3-LLM, the most powerful model ever built, was developed in 2020 using a two-phase technique that includes unsupervised pre-training and supervised tuning. As OpenAI continues to refine its GPT-4 model, there are significant improvements in the system's processing power and size [62]. LLMs frequently raise the issue of bias and unintended effects. These factors highlight several difficulties, including the size and structure of the model, the quantity and quality of the training dataset, and other problems [62, 63]. Performance depends on the tasks and environments in which they are utilized, and LLMs are currently being developed. Programmers create software with human-like thought and behavior patterns using ML and NN methods. Due to these techniques [64], they can offer enormous amounts of information to LLMs. Since the first applications of LLM techniques, NLP has made significant progress [65, 66]. These advancements have made computers better at understanding human speech. LLMs might produce an instructional resource by gathering textual information from many sources. A few applications that machine language can enable are chatbots, emotional detection, translation, summarization, and advances in various industries, including banking, healthcare, and education.

One advantage of LLM is that one can work in various language-related businesses without completing further training. However, they are less useful in data analysis and evaluation because of their associations with several problems, such as biases and ethics issues [67]. Information systems [68], which include a variety of sources such as Wikipedia, newspapers, documents, and social media, are frequently used as the subject of the LLM. Recognizing language patterns and relationships is essential for individuals to complete tasks, participate in conversations, and write effectively. The LLM training process usually takes weeks or months and requires computing resources such as efficient GPUs and sufficient memory. Language interpretation [69], chatbots [70], summarization [71], and cognitive reasoning [72] represent a group of many applications to which LLM can contribute. Moreover, this technology is also used to improve and speed up various language-related activities in fields such as banking [73], healthcare [74], and education [67]. LLM has many advantages, including the ability to perform NLP-related tasks without special training. It can be attributed to the experimental results because there are no clear instructions for performing specific tasks. In contrast, children can recognize and use many forms of conversation and cooperation [75, 76]. However, LLM shows some limitations and problems in its use, including ethics and injustice. In some cases, the scale and complexity of the LLM may create problems in understanding and evaluating the information contained, thus reducing its usefulness [67, 77]. Despite these difficulties, the LLM has become essential in artificial intelligence. These developments can change our understanding of language and how we interact with others. The constant evolution of the LLM will make it more attractive in many areas [78]. Over time, teachers have grown in size and complexity, allowing them to demonstrate an amazing ability to produce and understand language [79]. In addition, these techniques, such as language interpretation and question answering, are used in many NLP projects. However, as these structures become more complex, social and cultural problems such as unfairness and restrictions arise and require appropriate attention and solutions [76, 80]. Figure 10 illustrates different models of LLMs, and Table 7 presents a comparison of LLMs.

Models with multiple variables provide a deeper understanding of the connections between words and concepts [65, 82]. It is also recommended that the structure of the model be considered. Some jobs can be done more efficiently

Fig. 10 Training of LLMs

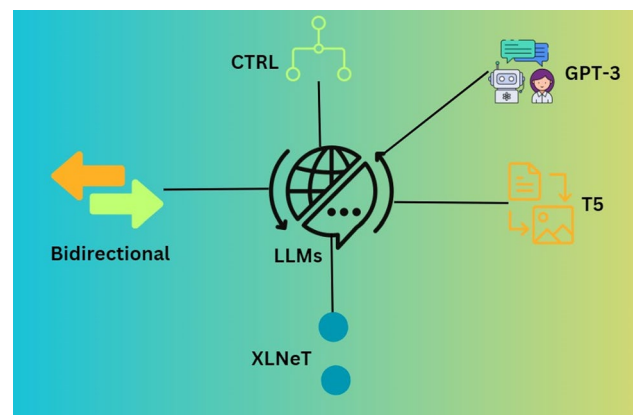


Table 7 Comparison of various GPTs

References	Model	Use	Objective	Year
[47]	GPT-1	General	12-level, 12-headed Transformer decoder (no encoder), followed by linear-softmax with Book Corpus: 4.5 GB of text	2018
[22]	GPT-2	General	GPT-1, but with modified normalization with Web Text: 40 GB of text	2019
[19]	GPT-3	General	GPT-2, but with modification to allow more extensive scaling with 570 GB plaintext	2020
[81]	InstructGPT	Conversation	GPT-3 fine-tuned to follow instructions using human feedback model	2022
[48]	ProtGPT2	Protein Sequences	As GPT-2 large (36 layers) with Protein sequences from UniRef50 of a total of 44.88 million	2022
[16]	BioGPT	Biomedical Content	As GPT-2 medium (24 layers, 16 heads) with non-empty items from Pub-Med total 1.5 million	2020
[17]	ChatGPT	Dialogue	UsesGPT-3.5, and finetuned with both supervised learning and reinforcement learning from human feedback (RLHF)	2022
[18]	GPT-4	General	Trained with both text prediction and RLHF and accepts both text and images as input, third-party data	2023

using one particular architectural style rather than another [83]. It is essential to consider the following parameters to evaluate LLM development effectiveness [84, 85].

- Size of training data.
- Good educational materials.
- Number of parameters
- Complexity of the model architecture.
- Task evaluation model.

It is also important to note that LLMs are still under development, and their performance can vary depending on the specific task and the environment in which they are used.

4.2 Training of LLMs

LLM must be trained to be successful. The first phase usually involves collecting and creating a large amount of information gathered from a variety of sources. The main source of LLM education is carefully reviewed and recorded [86]. Unsupervised learning is a method commonly used in educational institutions. The model is trained to use past reports as a base for predicting future reports in a given sequence. This phenomenon is widely recognized and is called "language modeling" [87]. Modern neural network architectures (such as Transformer) create relationships between words and sentences in a language [84, 88]. To increase the probability that the model will be selected in the next sentence in the context, the training process tries to identify negative aspects of the model [88]. It is common practice to use stochastic gradient descent (SGD) to achieve this goal. Backpropagation is then used to update the parameters of the model containing the back computation gradients. LLM learns from multiple objective feedbacks at every level of training. The first level of this study is initial training, which includes LLM training on a mix of unsigned and private objects without human supervision. The second stage is fine-tuning, which introduces narrower information and human feedback as access to simple models [90]. The fine-tuned model can go to another level where people use techniques to turn the LLM into an expanded model for doing the job.

4.2.1 GPT-3

One of the most advanced and well-known language models developed by OpenAI is GPT-3 [91]. This success generated significant interest and led to success in NLP [1]. The ability of GPT-3 to analyze complex networks and structures can be attributed to its foundation in transformer-based architecture [86]. The use of transformers makes GPT-3 possible. Due to its multilayer transformers, the model can understand and reduce text at many levels of abstraction. GPT-3 has an amazing 175 billion elements, making it one of the most comprehensive LMs ever created. GPT-3's training method mainly uses unsupervised learning using a lot of publicly available data. GPT-3 has a general understanding of English due to its small size and large training data. These systems can create cross-topic articles that display human-authored content. This work is done by optimizing the size of GPT-3.

4.2.2 Bidirectional

BERT, or "Encoder Representation of Transformer," is a popular LLM model for solving complex NLP problems. Pre-training, training, and modification represent different aspects of the training process [92]. In pre-learning, BERT identifies many redundant words to complete the representation. The tasks adopted in this study include multilayer masking (MLM) and next-sentence prediction (NSP). MLM improves the understanding of past and future content by clarifying a set of input tokens and training a model to predict the original tokens [93]. BERT influences NSP technology to improve understanding of relationships by analyzing whether the second expression follows the first expression. The BERT system uses data annotations to focus on specific tasks later in the initial training session. BERT's trained agents are designed for specific purposes, such as sentiment analysis and site recognition. Inverse processing and gradient descent are used to change the properties of the model. Field Programmable Gate Array (FPGA), Tensor Processing Unit (TPU), and Graphics Processing Unit (GPU) are required during BERT training [94]. The architecture of the BERT Transformer includes a sound input, residual connection, and normalization layer. This shows that the system can exhibit the ability to describe relationships and large-scale interactions between elements [95].

4.2.3 XLNet

An efficient method for resolving issues with conventional language modeling is to use the autoregressive pre-training technique that XLNet offers. Compared to traditional autoregressive models, XLNet is learned via switching [96]. Rather than relying on words found during training to generate predictions, XLNet analyzes all possible variants of the input and automatically predicts each variation's probability. XLNet overcomes the limitations of sequential models, which are limited to continuous evolution from left to right. This is achieved due to the ability to display relationships in both directions [79]. Unsupervised training is the first stage of XLNet training, and then there is a maintenance phase [97]. XLNet can predict a word by considering all input points without human supervision in the pre-training phase. The BERT model represents a change in design but contains significant changes. Using variables in XLNet training goals increases the complexity of some tasks. Compared to traditional auto-regressive models, XLNet requires a larger training size to cover more scenarios [98]. A set of multivariate auto regressive models can produce predictions using the underlying concepts of source and effect. Completing the training process on the computer increases costs and requires techniques such as factorization sampling to constrain the evaluation of the limited parameters of each training session. Another issue related to XLNet training is the huge demand for financial resources [99]. Due to its size and potential, it is essential to utilize additional memory and processing power. Training on XLNet requires multiple GPUs or TPUs, resulting in longer processing times [89].

4.2.4 T5

Developed and trained by Google, T5-LM is very easy to teach. During the training, T5 published a lot of content through public announcements on its website. The goal is to create a simple way to teach words that can be used in different contexts. The thing about T5 is that it knows all the hard work of creating text. Converting text from one format to another has been modified to simplify text-to-text conversion. Examples of these tasks include classifying text, writing, interpreting, and analyzing questions. For example, S5 does not need to give specific answers to the questions. However, users are advised to use all available information to complete the answer to the question. T5 was trained using a unique version of the Transformer architecture during initial training. Using the Transformer model [100] in T5 helps to accurately characterize the connection points at the entry, thus ensuring the stability of the connection between parts. The primary purpose of pre-training in maximum possibility estimation is to guide T5 to identify important text information based on the input [101]. After initial training, T5 will receive further modifications in subsequent tasks [61]. One of the challenges in T5 training is the limited availability of all the necessary documentation for the various applications. The effectiveness of this model is affected by the quality and quantity of data [90]. Gathering job-specific information is necessary for accurate correction. T5 training can require a significant investment of time and resources. The computational complexity of the model increases due to the presence of transformers and a large number of pre-learning parameters in the model architecture.

4.2.5 CTRL

CTRL-LLM can convert user instructions and control codes into text format. The training process for this model consists of two components: pre-training and validation. CTRL's initial training phase includes extensive knowledge that is freely available [102]. Pre-training aims to give the model the necessary skills to understand and generate meaningful textual content according to various control codes or signals [103]. Transformer designs similar to BERT [104] and GPT models are used for educational purposes. A model's predictive ability can be improved through training to analyze statistical patterns in language and grammatical structures to predict the following phrase in a given sentence. It is essential to perform this task in an organized environment. An important feature of CTRL is the ability to configure control messages and codes. The control code provided to the model acts as instructions and controls the text generation process [105] based on user-selected parameters, such as styles and other text components. The control code acts as a set of instructions that control the desired behavior of the model during both the training and inference processes [102]. The fine-tuning steps in the CTRL phase are the most important because they allow you to modify the model to adapt to a specific task or domain. The CTRL model is re-trained using a task-specific dataset

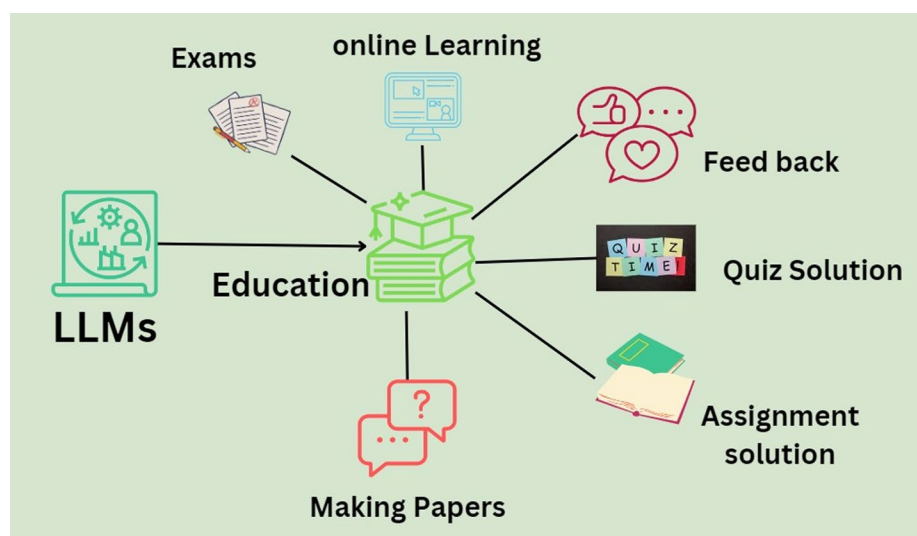
containing control code as part of the fine-tuning process. This model is trained to generate text that accurately represents the expected outcome after being provided with task-specific 531 inputs [106].

4.3 Role of LLM's in education

The impact of AI on institutions has been extensively examined in recent years. AI is essential in various domains, such as homework, assignments, course materials, and projects, and has seen significant changes following the release of ChatGPT [107] by OpenAI. During the conducted tests [108], it was observed that it showed an efficiency rate of less than 70%, indicating its potential failure to meet the requirements for AHA certification examinations. Nevertheless, the limited availability of a single response by ChatGPT resulted in a biased outcome, leading to a significant estimation of its capabilities. The way the investigation was conducted proved to be a critical flaw in judgment. The current accuracy of ChatGPT in BLS and ACLS is 96% and 92.1%, respectively.

Therefore, ChatGPT achieved good results in both tests, with 92 score. One of the advantages of integrating ChatGPT and AI agents into schools is their ability to get students to complete the task [109]. ChatGPT can save students time and energy by answering many questions while working. ChatGPT is recommended as it can provide guidance; bots can also play a role in completing the grading process, thus reducing teachers' workload and providing students with personalized and unique feedback. Another advantage of using ChatGPT and AI bots in the classroom is their ability to facilitate personalized learning. AI algorithms can analyze students' past performance on assignments and tests to provide recommendations for the future. Students can focus their efforts on these specific areas to maximize their performance [110]. Figure 11 shows the role of LLM in education. The non-profit organization, Khan Academy has expressed interest in exploring potential applications of ChatGPT within the framework of its daily operations. The technologically advanced trade machine "Khanmigo" with artificial intelligence has been successfully implemented. Digital formats of teaching materials are useful for classroom teachers [111]. ChatGPT can promote personal relationships between teachers and students, thereby improving the quality of teaching and education. AI can reduce misconceptions about its use in teaching and learning. Although AI is in the early stages of development, it can help students and meet their needs [112]. Using ChatGPT and AI class teachers may lead to negative consequences. Questions have been raised regarding the impact of these tools on children's cognitive abilities to think critically and creatively. Overreliance on AI robots to complete tasks and experiments can prevent the development of skills such as critical thinking and problem-solving, which are critical for success in the field of education. ChatGPT can provide valuable assistance in organizing classroom activities. The ability of AI chatbots to be an excellent tool for creating educational content is essential and can be explored in many ways. Course objectives, topics, syllabus, curriculum, assessment methods, learning engagement, and effective planning can be communicated quickly and effectively. Because ChatGPT provides feedback after each session, courses can be improved and modified based on input and observations from the model. The importance of ChatGPT as an additional tool that supports teachers' ability to support and educate students cannot be ignored. The collaboration of humans and AI in curriculum design is very beneficial as it reduces the creation of efficient and effective.

Fig. 11 Role of LLMs in education



4.4 Role of LLM's in higher education

The use of LLM can help professionals, students and teachers in schools. Seven options are available for the LLMs. Two programs are designed for professionals and teachers, and the remaining five are designed for students. The LLM understands itself as a tool that provides personalized lessons, explanations, and advice personalized to students' specific questions to help them learn independently. According to statistics, LLM can be considered an effective teaching tool that provides personal feedback on students' writing and encourages discussion by encouraging individuals to share their thoughts. Research on the previously mentioned subject backs up this claim [113]. Similarly, besides the usual classroom equipment, LLM will be an additional tool to improve students' knowledge retention and thinking skills. Therefore, research LLM can be an additional tool to provide theoretical answers and formulate published recommendations [114]. These components continue the work of the LLM and include communication skills, language therapy, and writing and language support. Incorporating these tools into the LLM can reduce enrollment by reducing barriers to entry into higher education.

The authors of [115] explain that LLM translation and remediation programs can increase the integrity of education by supporting students who do not speak English. This is especially helpful for students whose families lack English skills. Using appropriate keywords and descriptions will help organize the content to ensure no spelling or grammatical errors. Figure 12 illustrates the role of LLMs in students learning.

Also, LLM agrees to maintain ongoing support for students. People participating in remote or international research in countries with different opportunities may find this a helpful tool [67]. In existing studies, little attention is paid to the benefits of the LLM. Integrated LLM into the curriculum can bring new, interactive learning to the benefit of students. Although possible, some exciting documentation exists on using LLM in developing digital learning ecosystems [116].

Initially, the tool was designed to assist teachers and educators by providing timely advice, answers to everyday questions, and guidance on frequently asked questions. Based on existing research, the LLM has many benefits for teachers. In particular, it provides personalized learning support by allowing students to adapt resources and activities to meet their needs. Also, teachers can use LLM to match course objectives, learning outcomes, and standards [117]. Figure 13 illustrates the opportunities for using LLMs in higher education.

4.4.1 Opportunities

LLM has been used to create personalised learning materials since its development. Although it is still early, the above conclusions appear mostly theoretical or hypothetical. Until further empirical studies confirm the validity of these concepts, attention is advised in current practices. This strategy must be tested in the real world to verify its effectiveness. Finally, LLM students say they can contribute to science education through their ability to help generate ideas, analyze data, and conduct reviews. This approach is supported by previous research showing that using LLM can improve the writing of training [115], promote collaboration, facilitate the organization of activities, and develop an environment helpful to creating good results [118]. Despite this concern, the negative impact of education on self-efficacy and independence is still a concern [119]. More education is needed on intellectual skills in research and writing. However, some authors

Fig. 12 LLM's in students learning

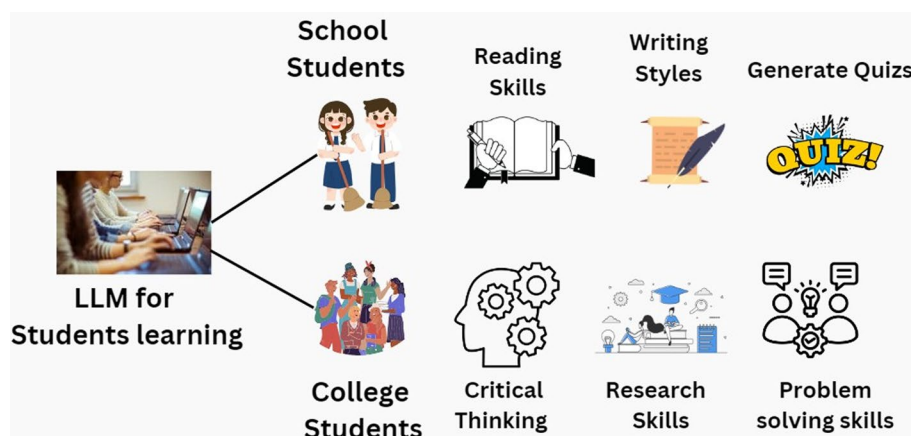
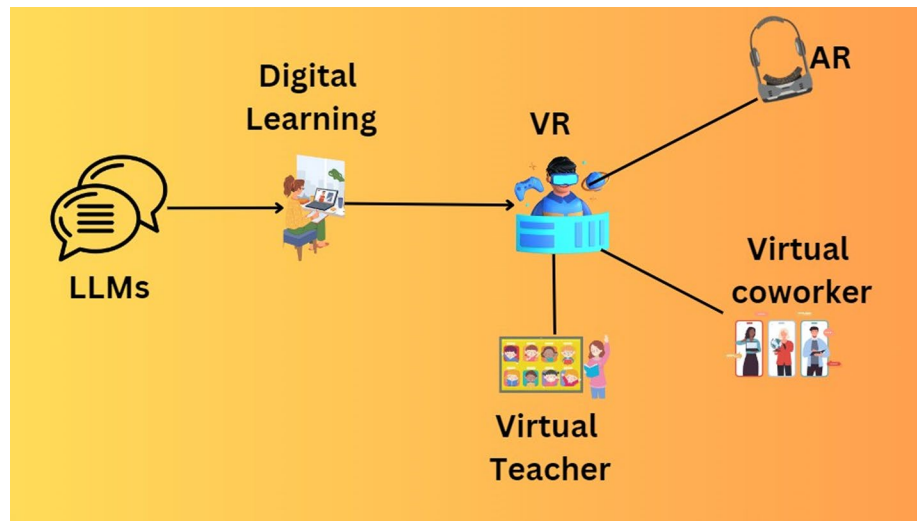


Fig. 13 Role of LLMs in digital education



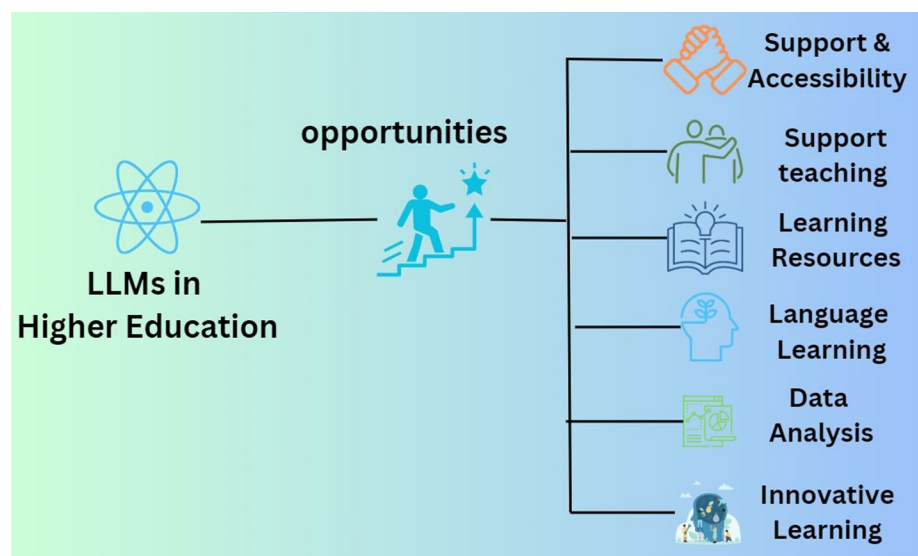
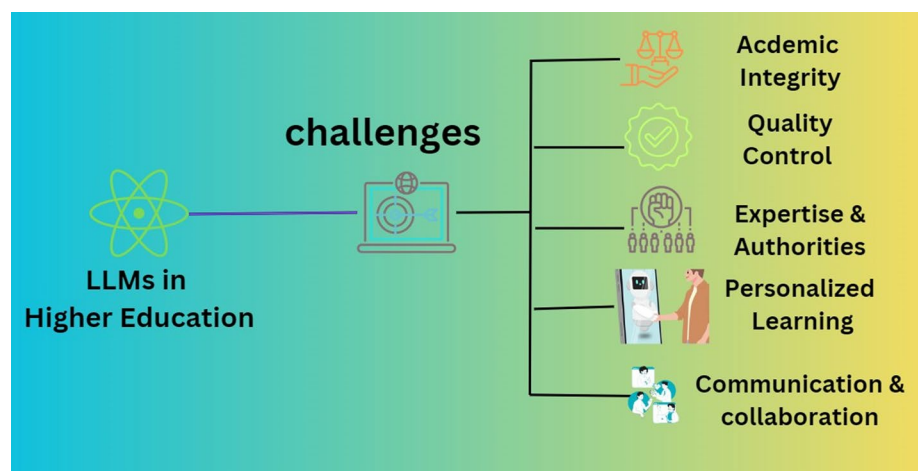
have begun to give general advice about the potential benefits of the LLM, which can lead to the writing process [27]. According to [119], it is essential to manage the risks sensibly, leading to research and significant results such as inquiry, creativity, and discovery. However, the authors also emphasize the importance of ensuring the best results from AI. The authors recognize that providing a link between the necessary human resources—education, training, and creativity—is essential for advancing scientific knowledge.

4.4.2 Challenges

LLM inaccuracy and lack of trust in data produced by artificial intelligence such as GenAI and its impact on higher education. Another question arises about how students use it. The teachers of Law face four main challenges: quality control, information and policy, formal education and communication, and collaboration. LLM strives to provide accurate and helpful information; however, sometimes, it can give wrong or incorrect answers. It demonstrates the potential impact of bias in LLM and other GenAI systems on the reliability and quality of the information they provide [120]. For the benefit of all, the LLM recommends ongoing monitoring, bias awareness, and ongoing training in common sense skills as strategies to further reduce bias. LLM students often understand the limitations of their specific knowledge and experience provided by professionally trained professors and teachers.

University education is limited to LLM and cannot provide students with depth of knowledge, rigorous analysis, and focused learning. According to LLM, it is recommended that schools emphasize the role of teachers and teaching staff in guiding students, and ensuring. Figure 14 shows the LLM's opportunities in higher education.

The knowledge is built from intelligence, and providing necessary background knowledge, especially regarding quality control [121]. Previous research has not paid sufficient attention to two critical aspects: self-directed learning, communication, and collaboration. Despite initial comments regarding its ability to enhance personal learning [122], the LLM is banned but has not yet been discovered. LLM only hopes for a customized education; some aspects of students' needs, such as their learning preferences, standards, and specific problems, may be ignored. This cannot communicate effectively about the small characteristics that distinguish one student from another. This example demonstrates the importance of key faculty members in monitoring the use and implementation of LLMs in an academic environment. The third problem is the barrier to cooperation and communication between university students. The need for collaboration in group work and other learning activities will reduce their overall effectiveness. Individual responses to the teachers show that although technology has its advantages, it has not yet reached a level that could fundamentally change the human condition. For this reason, those who want to use it in education should provide the necessary training to teachers to use it effectively and reduce the harmful effects listed above. LLM raises another concern regarding the abuse of their powers, particularly their lack of integrity in education. According to the master's thesis, students may use the model to copy content from other sources or engage in dishonest learning by cheating on assignments and exams. Whether universities should ban this technology entirely is a matter of ongoing debate, mainly because it has many aspects in the educational process [123]. Figure 15 shows the challenges faced by higher education using LLM Plagiarism and other forms of academic dishonesty are problems in higher education and have led to extensive research on this topic [124].

Fig. 14 LLM's opportunities in higher education**Fig. 15** Challenges in higher education using LLM

However, due to the novelty of ChatGPT and the lack of working methods, the risk associated with ChatGPT is high. Recent research shows that schools may not know how to manage the best strategies for students using AI applications. It is observed that from 142 schools reviewed in May 2023, only one was found to have implemented a policy prohibiting the use of artificial intelligence [125]. This decision is important because it contradicts previous research [126] showing that learning unfairness is reduced when people are aware of and follow the principles of fairness theory. This demonstrates the potential impact of following academic integrity principles in reducing cheating in education.

Universities must establish a clear policy to prepare them to use the Master's degree as a teaching method and assessment tool. The need to update the code of ethics stops the need to resolve problems caused by Gen-AI tools. Students need to evaluate the changes in the use of artificial intelligence technology and the consequences that may arise if the content they use is not followed [125]. The nature of artificial intelligence creates severe challenges in developing general laws to address the changing problems arising from artificial intelligence. Consistent with existing research, it has been shown that students are more likely to engage in immoral behavior, such as lying, when they see that they are less likely to be detected [126]. Therefore, a cultural environment that rejects unfair learning can be fostered through the use of GenAI.

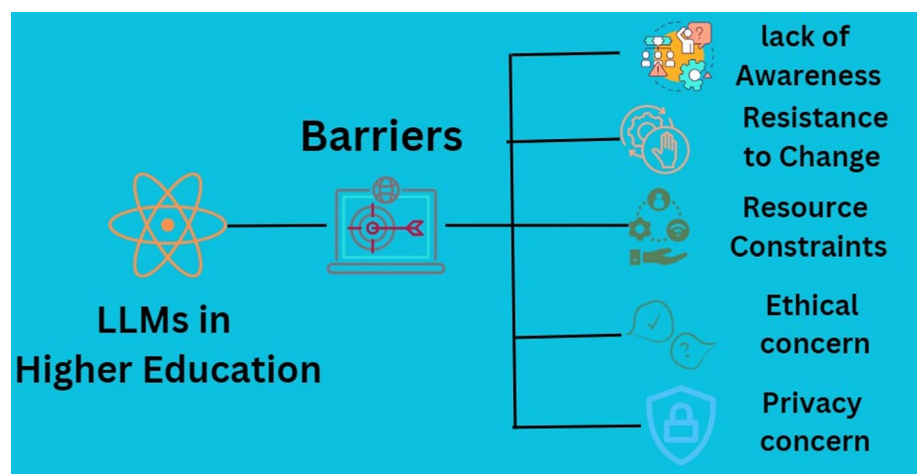
4.4.3 Barriers

LLM analyzed the challenges that universities face when using GenAI technology. The impact of factors such as ethical and privacy concerns, collaborative governance, resource constraints, complex legal and regulatory environment navigation, and general misunderstanding is significant. LLM provided valuable insight into the challenges

universities may face during deployment [127]. However, many of these challenges remain unexplored, providing time for further analysis. These measures should improve teachers' and students' understanding of integrating ethical AI into education. Many of the challenges to be aware of include confidentiality, ethical considerations, and the quality and consistency of the study. According to [127, 128], expanding the right to self-determination should take into account how crucial it is to redefine what responsibility means. Responsibility and control of intellectual property ultimately rest with the user. Therefore, it is essential to conduct timely research on intelligence ethics in education. Ultimately, the outcome of this effort should be the eventual establishment of laws regarding the fair use of intellectual property in schools. Therefore, more research on the ethics of quick intellectual property rights in education is urgently needed, as are efforts to promote the exchange of information and provide rapid training [129]. Schools with limited budgets or short staff might face challenges in taking advantage of the resources offered through ChatGPT. Similarly, using Gen-AI will require new tools, such as efficient processing systems and reliable internet connections. This means that universities that do not have sufficient resources cannot benefit from the advantages of LLM. Although universities have the opportunity to do this, they may still face resistance when implementing changes. Employers, managers, and academic staff may prefer traditional courses. Recent research suggests that ChatGPT in classroom planning and assessment can yield significant improvements [130].

Additionally, efforts will be made to provide training and facilitate information sharing among key stakeholders, such as teachers and students, to improve their understanding and knowledge of female students. According to LLM's review, schools with limited financial resources or inadequate staff may face difficulties using the current system. Similarly, effective use of Gen-AI technology will require robust network connectivity and high-performance hardware. This means that universities that do not have the necessary resources cannot benefit from the quality of the LLM. It was thought that universities could be successful with financial constraints by acting unfairly. In this case, they will encounter resistance when using the changes. Employees, managers, and professionals may prefer traditional methods or resist integrating ChatGPT into their operations. Recent research shows that using ChatGPT in lesson planning and evaluation can increase effectiveness [130]. As we increase the use of ChatGPT in educational environments, publishing research studies on ChatGPT is essential. A limitation is that proving its quality may lead to more problems in the real world. According to a recent study, it was found that students have a positive attitude towards Gen-AI technology and consider using it regularly as a habit pattern [127]. Most of the research focuses on the LLMs, an Internet-based platform designed to enable students to engage in peer-to-peer discussions beyond the boundaries of traditional education. Universities, therefore, have a unique opportunity to reflect on the ethical foundations of AI research and teach students behaviors that support these values. This is important to ensure that AI models such as LLM promote ethical standards, protect student rights, and enhance educational opportunities. Figure 16 shows the constraints in higher education.

Fig. 16 Constraints in higher education



4.5 Key challenges and risks related to the application of LLMs in education

4.5.1 Difficulty to distinguish model-generated from student-generated answers

Teachers now have a particular job to find who wrote something on paper. They must deal with growing difficulty in telling if it was made by a computer or human [131]. So, the New York City Department of Education stopped using ChatGPT. They won't let it into schools or networks anymore [132]. The writer of [131] has suggested many ways to find and sort text made by LLMs, like ChatGPT. People will also expect how helpful new tools like GPTZero can be. This tool uses confusion to find out if the language came from AI. It's a way of measuring how good an agent is at understanding different things. A common goal of modern methods is to use watermarks for content created by LMs [133]. A way to do this is by pushing material that uses unusual mixes of words.

4.5.2 Cost of training and maintenance

Some educational institutions may have financial constraints that prevent them from administering the existing LLMs [67]. A potential way to overcome this challenge is through the joint use of advanced education models, cloud technology, and recommendations developed in collaboration with businesses and schools.

4.5.3 Data privacy and security

The use of LLMs Changes in the learning environment raises concerns about the protection and privacy of student information [134]. This is because student information is generally considered confidential. The report addresses issues such as data breaches, un-authorised access to student information, or misuse of student information for financial growth [133, 134].

4.5.4 Sustainable usage

To make great efforts, spending a lot of money is necessary. Schools must use energy-efficient technologies to ensure environmental sustainability and integrate systems based on renewable energy sources [135], such as cloud computing. This approach is essential for the school's success and future growth [136]. Standard training and correction materials should be limited to legal, ethical, and adequate information. Therefore, a management structure that includes policies, procedures, and controls must be in place to ensure that the implementation of these standards is successful. Likewise, for the model to be used in a reliable, ethical, and stable manner in the long term, it must have features such as openness, objectivity, and continuous evaluation.

4.5.5 Cost to verify the information and maintain the integrity

The data model must be monitored with approval from a reliable external source to ensure the originality and accuracy of the model. Also, consider the costs associated with regularly updating and maintaining the model to ensure it always provides accurate results. This protection is taken to ensure that the model performs its planned function.

4.5.6 Difficulty in distinguishing between actual writing and model output

The writing ability of the LLM resembles human writing, which allows students to distinguish between suggestions and unreliable information. As a result, students are more likely to accept false or misleading information without inquiring about its accuracy. In addition to recent debates about the accuracy and completeness of data, student's ways of researching, examining, and evaluating data for identification, verification, and uttering their opinions are essential for reducing the risk of this gender.

4.5.7 Lack of adaptability

Because master's degrees cannot meet different needs, they may not provide students and faculty with the flexibility needed for effective learning. Future data may be more flexible, although current technology prevents them from being flexible [137]. Figure 17 shows the main challenges of LLMs in education.

4.6 Solution of challenges

4.6.1 Teachers may rely too heavily on the model solution

There are many ways to solve these problems.

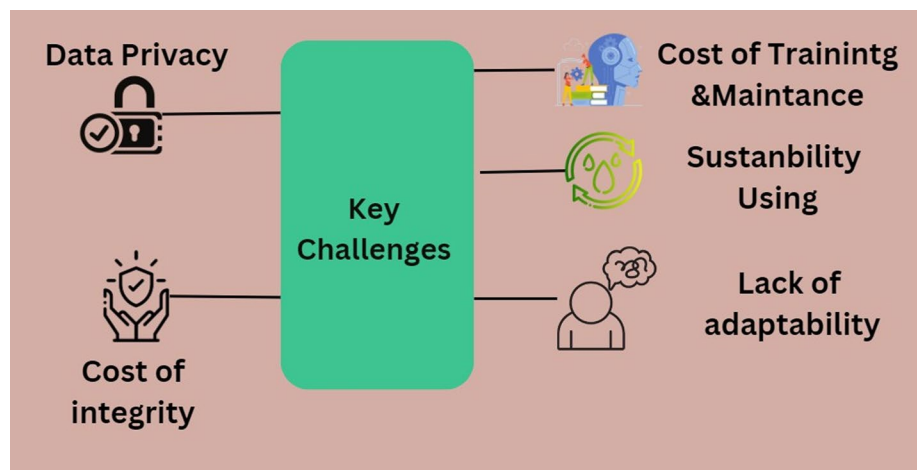
- Teachers have access to modern technology that will encourage children to think and create original solutions.
- Educational activities that use technology to develop children's thinking and problem-solving skills. Students should participate in independent, creative activities that allow them to develop and apply their ideas and methods.
- Monitor the use of LLMs in the classroom through assessment and evaluation to ensure they do not interfere with student learning.
- To encourage teachers and schools to develop the teaching of group work. It involves using models and other methods to find and evaluate the information needed for a resource project.

4.6.2 Lack of understanding and expertise solution

Below are different approaches to addressing these issues and accepting this risk.

- LLMs in educational institutions that require assessment and case-based support for teachers and students to use the LLM safely and ethically. The introduction and use of MLM in the classroom should be introduced to teachers and schools through training and continuing education. Training and continuing education programs allow educators to learn more about the benefits and uses of LLMs in the classroom.
- Teachers and schools can learn more about using LLMs in the classroom using open educational resources such as tutorials, research, use cases, etc.
- Encourages educators and teachers to participate in active groups (such as professional learning communities). LLMs allow them to share resources and benefit from each other.
- LLMs require continuous analysis and feedback on their applications to ensure correct use and change courses as necessary.

Fig. 17 Key challenges



4.6.3 Difficulty to distinguish model-generated from student-generated answers solution

Therefore, to ease the impact of this situation, you should prioritize the following in your strategy:

- Professionals must review issues such as transparency, interpretation, measurement, and analysis to differentiate between human and computer-generated data.
- To support and encourage the creation of courses and instruction for the new and effective use of teachers.

4.6.4 Cost of training and maintenance solution

The following points should be considered when trying to reduce this risk. The graphical representation of the solution is illustrated in Fig. 18:

- It uses previously studied, publicly available models that can be adapted to a variety of specifications.
- The aim is to create and explore partnerships with various organizations, including businesses, schools, governments, and charities. These collaborations focus on financial support, resources, and expertise to facilitate the implementation of LLMs in education.
- Use scalable computing services (such as cloud)

4.6.5 Data privacy and security solution

The following points are needed to solve this issue:

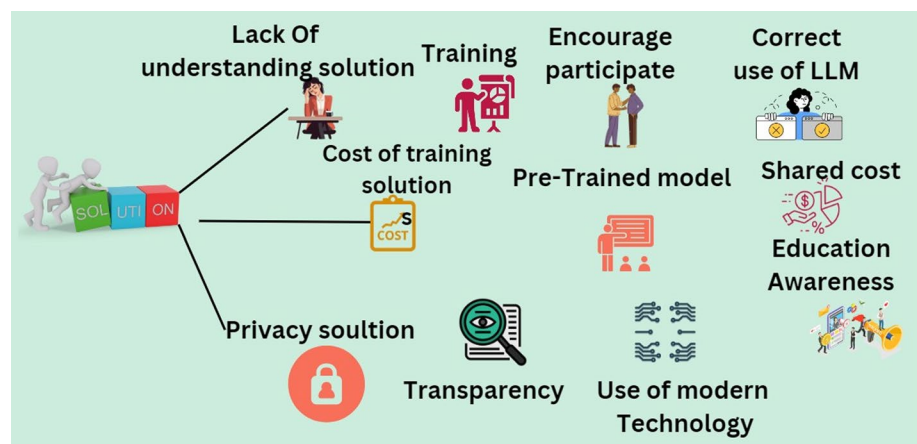
- The General Data Protection Regulation, the Health Insurance Portability and Accountability Act, and the Family Educational Rights and Privacy Act are just a few examples of the legal and moral guidelines that must be followed while gathering, storing, and using student data.
- The students' relatives must communicate and accept the rules regarding collecting, storing, and using student data.
- Privacy-protecting analysis approaches, current encryption and federation techniques, and other measures should be implemented to prevent hacking, unauthorized access, and unethical use of gathered data.

4.6.6 Sustainable usage

Some examples of materials and equipment that improve energy use are.

- Hardware and electrical equipment that use renewable energy, low training, and research to reduce maintenance costs (such as simplified algorithms, data representation, and storage),
- Ethics and exposes, processing and storage, evaluating legal documents and identifying bias, and using transparent and interpretive processes to prevent bias.

Fig. 18 Solution of challenges



4.6.7 Cost to verify the information and maintain the integrity of the solution

When addressing this risk, a risk management plan should include the following:

- The model must be regularly updated with new written information to ensure consistency in producing current and accurate results. Ensure the accuracy and reliability of data by verifying data with multiple reliable sources. Improve assessment and authorization by combining the insights and expertise of teachers or experts with established standards.
- The user points out the development of a combined process and method for evaluating the accuracy of predictive models [138].
- Provide clear and precise information about the model's performance, highlight its strengths and weaknesses, and identify insights and practical applications.
- Regular monitoring is essential. It is important to identify errors and other potential problems. These resources can help teachers and students implement strategies, understand research results, and evaluate the collected data [138].

4.6.8 Lack of adaptability solution

The following steps are taken to overcome this issue:

- Adaptive learning technologies influence student data, including their learning styles and prior knowledge, to personalize the model's output to provide for the unique needs of each student.
- The customization of learning and teaching extends to various media, including text, audio, video, and practical experimentation, ensuring engagement and effectiveness.
- Customization of the language model's output is achievable by aligning it with the instructor's teaching style and the specific course topic, utilizing insights provided by the teacher.
- Learning materials that are both individualized and targeted become feasible through hybrid approaches. These approaches merge the advantages of human instructors with those of language models, taking into account teachers' feedback, guidance, and support.
- Regular evaluation and standard development are essential, especially in classroom applications. This ensures that the model delivers an accurate and efficient learning experience.
- Continuous research and development aims to create more flexible models that meet the needs of students and teachers. This constant change is designed to improve learning skills.

5 Discussion

The integration of LLMs, such as ChatGPT, into education represents a transformative opportunity but also raises multifaceted challenges that must be carefully considered. This section provides a critical analysis of the benefits, risks, and actionable strategies related to LLM deployment, incorporating ethical dilemmas, cultural considerations, and pedagogical impacts to ensure a balanced perspective.

5.1 Opportunities for education stakeholders

The opportunities provided by LLMs extend to educators, students, institutions, and policymakers, with far-reaching implications for improving educational systems:

- For Educators:
 - Enhanced Curriculum Development: LLMs streamline curriculum creation by generating tailored content, exercises, and lesson plans aligned with diverse student needs [34]. This allows educators to focus on higher-order tasks, such as fostering critical thinking.
 - Time Efficiency through Automation: Automating grading and feedback processes enables teachers to dedicate more time to mentorship, classroom interaction, and personalized guidance.

- Personalized Tutoring: LLM-powered tools support real-time, adaptive tutoring for individual learners, addressing gaps in knowledge and engagement. However, it remains critical that educators oversee these tools to ensure pedagogical alignment.
- For Students:
 - Individualized Learning Paths: By adapting to students' learning paces, styles, and preferences, LLMs facilitate inclusive and personalized learning experiences [9]. This is particularly valuable for students in underserved or remote areas.
 - Language and Accessibility Support: Tools such as real-time translators and speech recognition systems improve access for non-native speakers and students with disabilities, fostering equity in education [33].
 - Improved Engagement: Intelligent tutoring systems and AI-powered simulations promote interactive and immersive learning experiences, increasing motivation and retention.
- For Policymakers and Institutions:
 - Administrative Optimization: LLMs automate processes like student performance analysis, feedback aggregation, and course recommendations, enabling data-driven decision-making.
 - Policy Development: Policymakers can leverage AI insights to create policies addressing digital education challenges, such as accessibility, data privacy, and ethical use.
 - Cost Efficiency: Institutions can reduce operational costs through AI automation while enhancing resource allocation for targeted improvements.

5.2 Challenges and mitigation strategies

Despite their promise, LLMs present critical challenges that must be addressed to ensure ethical, responsible, and effective deployment in education.

5.2.1 Ethical and pedagogical considerations

- Overreliance on AI Tools: Excessive dependence on LLMs may undermine teachers' roles as mentors and students' development of critical thinking and problem-solving skills [8, 38].

Mitigation: Encourage teacher-AI collaboration where educators use LLMs as assistive tools rather than replacements. Professional development programs can train teachers to integrate AI responsibly.

- Bias in AI Outputs: LLMs may reflect societal, cultural, or linguistic biases present in training datasets, leading to discriminatory outcomes [21, 37].

Mitigation: Implement bias detection and mitigation frameworks, such as regular audits, diverse training datasets, and algorithm transparency, to ensure fairness and inclusivity.

While previous studies [21, 37] have underscored the risks of algorithmic bias in AI-driven tools, our findings contribute to this discussion by emphasizing the importance of culturally diverse datasets and proposing specific mitigation strategies, such as bias audits and inclusive training frameworks.

- Cultural Sensitivity: LLMs risk perpetuating dominant cultural norms at the expense of local or marginalized perspectives.

Mitigation: Promote the development of context-aware AI tools that accommodate linguistic diversity and cultural nuances to foster inclusive education.

5.2.2 Data privacy and security

Handling sensitive student data raises concerns about privacy breaches and misuse. Institutions must:

- Adopt robust encryption and anonymization techniques to protect personal data.
- Ensure compliance with global regulations (e.g., GDPR, FERPA) through clear governance policies.

- Provide regular training for educators and administrators on privacy standards.

5.2.3 Sustainability and environmental concerns

LLM deployment consumes substantial energy, contributing to environmental impact. Strategies to address this include:

- Utilizing energy-efficient AI models optimized for educational use.
- Collaborating with technology providers to integrate renewable energy solutions for data centers.
- Supporting research into sustainable scaling practices for AI technologies.

5.2.4 Regulatory and governance challenges

The lack of clear governance frameworks complicates the ethical and legal implementation of LLMs in education. Institutions must:

- Collaborate with policymakers and legal experts to develop regulatory guidelines for AI use.
- Establish ethical oversight committees to monitor LLM deployments and ensure transparency.

5.3 Critical pedagogical impacts

LLMs have far-reaching implications for pedagogical practices. While they enhance automation and personalization, they also require critical reflection on their role in teaching and learning:

- **Maintaining Human-Centric Learning:** Teachers remain indispensable in fostering creativity, emotional intelligence, and higher-order thinking, which cannot be replaced by AI tools.
- **Promoting Critical Thinking:** LLMs should supplement teaching by enabling students to question and evaluate AI-generated outputs, developing critical digital literacy.
- **Balancing Equity and Access:** Institutions must ensure that LLM deployment does not widen the digital divide by promoting accessibility for all learners, particularly in low-resource environments [21].

5.4 Practical recommendations

To ensure the responsible and effective use of LLMs, this study proposes actionable recommendations for key stakeholders:

For educators:

- Integrate LLMs as tools for lesson planning, content personalization, and automated feedback while preserving human oversight.
- Participate in professional development programs to learn effective collaboration with AI systems.

For policymakers:

- Develop regulatory frameworks for ethical AI deployment, data privacy, and fairness.
- Provide funding for research into energy-efficient, inclusive, and bias-aware AI models.

For researchers:

- Explore bias detection algorithms and strategies for fairness in LLMs.
- Develop energy-efficient AI architectures to minimize environmental impact.
- Investigate contextual adaptations of LLMs to ensure cultural and pedagogical relevance.

5.5 Summary of key insights

The integration of LLMs in education offers significant opportunities, including enhanced personalization, administrative optimization, and accessibility improvements. However, the deployment of these tools raises complex ethical, cultural, and pedagogical challenges that require proactive strategies.

By addressing data privacy, mitigating biases, promoting human-AI collaboration, and supporting sustainability, LLMs can be effectively integrated into education to create a balanced, inclusive, and adaptive learning environment.

This study contributes to the existing literature by not only identifying the challenges and opportunities of LLMs in education but also proposing a structured theoretical framework and actionable strategies that address underexplored areas such as ethical dilemmas, cultural inclusivity, and the role of LLMs in specific learner populations. These contributions provide a foundation for future research and real-world implementation.

6 Conclusion

6.1 Recapitulation

This study explored the transformative potential of LLMs like ChatGPT in education, highlighting their role in enhancing personalized learning, supporting educators, and improving accessibility. The article proposed a novel theoretical framework for integrating LLMs in educational settings, focusing on personalized learning models, ethical considerations, and adaptability across diverse contexts.

While LLMs offer significant opportunities, the study also identified key challenges, including privacy concerns, ethical biases, energy consumption, and regulatory compliance. By adopting ethical deployment practices, sustainable scaling solutions, and teacher-AI collaboration strategies, these challenges can be mitigated.

6.2 Future directions

To ensure the responsible and effective integration of LLMs into education, future research should address the following key areas:

- Best Practices for Integration
 - Develop comprehensive frameworks to guide educators on the optimal use of LLMs in classrooms while aligning their deployment with learning objectives and curricula.
 - Explore methods for integrating LLMs into blended learning environments, ensuring they complement human-led instruction without compromising pedagogical standards.
 - Investigate effective training programs for educators, enabling them to leverage LLMs to enhance teaching quality and student engagement.
- Bias Detection and Mitigation
 - Design and implement bias-detection algorithms to monitor and reduce biases in LLM outputs, particularly in diverse educational contexts.
 - Explore techniques for creating fair and diverse training datasets to minimize systemic biases and ensure equity across student groups.
 - Study the impact of bias on educational outcomes and develop strategies to address unintended consequences in LLM-driven applications.
- Privacy and Security Enhancements

- Investigate advanced data encryption techniques and anonymization protocols to safeguard sensitive student information during LLM deployment.
 - Develop privacy-preserving frameworks that comply with evolving data protection regulations (e.g., GDPR, FERPA).
 - Study the risks associated with data storage and sharing in AI-powered educational tools and propose secure implementation practices for institutions.
- **Energy Efficiency and Sustainability**
 - Promote research on low-energy LLM architectures optimized for educational tasks, reducing computational and environmental costs.
 - Explore the integration of renewable energy solutions and energy-efficient hardware for AI infrastructure in educational settings.
 - Develop frameworks for sustainable AI scaling, ensuring that LLM adoption aligns with global environmental priorities.
- **Human-Centered AI Collaboration**
 - Study how LLMs can best support teachers in curriculum design, content generation, assessment automation, and personalized student feedback.
 - Investigate strategies to maintain the critical role of teachers in fostering human interaction, mentorship, and critical thinking.
 - Examine the professional development needs of educators to ensure effective collaboration between human teachers and AI tools.
- **Regulatory and Ethical Compliance**
 - Focus on developing clear guidelines for policymakers to regulate LLM use in educational institutions, including ethical AI deployment, accessibility standards, and legal compliance.
 - Explore frameworks to ensure transparency and accountability in LLM implementation, addressing concerns about fairness, bias, and ethical use.
 - Promote cross-disciplinary collaboration among educators, policymakers, and legal experts to address regulatory challenges.

By addressing these future directions, researchers, educators, and policymakers can ensure the responsible, ethical, and sustainable integration of LLMs in education. These efforts will enable LLMs to enhance learning experiences, empower teachers, and address systemic challenges while minimizing potential risks and unintended consequences.

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Declarations

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